TRIAGEFLOW
INDUSTRIAL EQUIPMENT MANUAL

Maintenance & Troubleshooting Reference — Volume 1

Document ID: TF-MAN-001
Revision: 2.1
Classification: Internal Use — Synthetic Training Data
Prepared for: TriageFlow RAG Knowledge Base (Vector Store Ingestion)

How This Manual Is Organized

Section	Equipment Category	Equipment IDs Covered
1.x	Centrifugal Pumps	P-204, P-207, P-318
2.x	Reciprocating & Screw Compressors	C-11, C-22, C-305
3.x	Electric Motors & Drives	M-18, M-44, M-091
4.x	Steam & Gas Turbines	T-501, T-512
5.x	Valves & Pressure Relief Systems	V-118, V-203
6.x	Heat Exchangers	HX-77, HX-88
7.x	Electrical Switchgear & MCCs	MCC-14, SW-29
8.x	General Safety Override Reference	—

Document Purpose

This manual is a synthetic, fictional reference document created specifically for the TriageFlow project's retrieval-augmented generation (RAG) pipeline. It is modeled on the structure, terminology, and equipment categories used in real industrial facilities (rotating equipment, pressure vessels, electrical systems) but all equipment IDs, specifications, and incident histories are fabricated for testing purposes only.

This document is designed to be:

- **Chunked** by backend/services/vector_store.py using recursive character splitting
- **Embedded** into ChromaDB for semantic search
- **Retrieved** by backend/agents/retrieval_agent.py in response to maintenance tickets
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Each section below uses consistent numbering (Section X.Y) so retrieved chunks carry meaningful citation metadata.

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SECTION 1: CENTRIFUGAL PUMPS

1.1 Equipment Overview — Centrifugal Pump Class

Centrifugal pumps use a rotating impeller to convert mechanical energy into fluid velocity and pressure. They are the most common rotating equipment type in liquid-handling services across upstream, midstream, and downstream facilities. Common failure modes include bearing wear, seal degradation, cavitation, misalignment, and impeller damage from particulate ingress.

1.2 Equipment Record: Pump P-204

Equipment Type: Horizontal Centrifugal Pump
Service: Produced Water Transfer
Rated Flow: 450 m^3/h
Rated Head: 85 m
Driver: Electric Motor (M-18, see Section 3.2)

1.2.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Increasing vibration, steady temperature	Bearing wear (early stage)	Medium — schedule within 5-7 days
Increasing vibration AND rising temperature	Bearing wear (advanced) or misalignment	High — schedule within 24-48 hours
Loud knocking/rattling noise	Cavitation or impeller damage	High — investigate immediately
Sudden loss of discharge pressure	Impeller wear, seal failure, or suction blockage	Critical if accompanied by leakage
Visible leakage at seal	Mechanical seal failure	High — risk of process fluid release
Leakage with unusual odor or vapor cloud	Possible hazardous fluid release	Critical — evacuate area, treat as emergency

1.2.2 Troubleshooting Procedure — Vibration Increase

1. Confirm vibration reading against baseline using handheld vibration meter at bearing housing (inboard and outboard).
2. Check lubrication levels and lubricant condition (look for discoloration indicating water ingress or overheating).
3. Inspect coupling alignment between P-204 and M-18 driver motor.
4. If vibration exceeds 7.1 mm/s RMS (ISO 10816 Zone C boundary for this pump class), schedule bearing inspection within 48 hours.
5. If vibration exceeds 11 mm/s RMS (Zone D), shut down pump and inspect immediately — continued operation risks catastrophic bearing failure.
6. Document all readings in the equipment health log before closing the work order.

1.2.3 Required Parts (Typical Bearing Replacement)

- Bearing set, P-204 inboard/outboard (Part No. BRG-204-IB / BRG-204-OB)
- Mechanical seal kit (Part No. SEAL-204-K2)
- Coupling alignment shim set
- Estimated labor: 4-6 hours, 2 technicians

1.2.4 Escalation Rule

If a vibration-related ticket for P-204 also mentions **leakage**, **unusual smell**, or **smoke**, this must be escalated to Critical regardless of vibration reading — produced water transfer lines at this facility carry trace H2S, and any uncontrolled release is a safety event, not just a mechanical one.

1.3 Equipment Record: Pump P-207

Equipment Type: Vertical Centrifugal Pump
Service: Cooling Water Circulation
Rated Flow: 1,200 m^3/h
Rated Head: 40 m
Driver: Electric Motor (M-44, see Section 3.3)

1.3.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Reduced flow output, steady pressure gauge	Strainer blockage upstream	Low-Medium — schedule cleaning
Pump running but no flow	Air lock or dry running	High — stop pump to prevent seal damage
Intermittent vibration tied to flow rate changes	Cavitation due to low suction head	Medium — review suction conditions
Bearing housing hot to touch, vibration normal	Lubrication failure	High — risk of imminent bearing seizure

1.3.2 Troubleshooting Procedure — Loss of Flow

1. Verify suction strainer differential pressure; if above 0.5 bar, strainer is likely blocked.
2. Check suction valve position — confirm fully open.
3. Verify pump casing is fully primed (vertical pumps are prone to air lock after maintenance).
4. If dry-running is suspected, do NOT restart the pump — dry running for more than 30 seconds on this model risks seal face cracking.
5. Once flow is restored, monitor for 15 minutes before closing the ticket.

1.3.3 Required Parts

- Strainer basket replacement (Part No. STR-207-B)
- Mechanical seal kit if dry-run damage confirmed (Part No. SEAL-207-K1)

1.4 Equipment Record: Pump P-318

Equipment Type: Multistage Centrifugal Pump
Service: Crude Oil Booster (High Pressure)
Rated Flow: 280 m^3/h
Rated Head: 620 m (multistage, 5 stages)
Driver: Electric Motor (M-091, see Section 3.4)

1.4.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Discharge pressure fluctuating ±10%	Internal stage wear or recirculation	Medium
Sudden pressure spike with audible bang	Surge event / valve slam	Critical — possible pipe/seal damage
Visible crude oil seepage at any flange	Gasket or seal degradation	Critical — hydrocarbon release risk
High-pitched whine increasing in pitch	Bearing or impeller cavitation damage progressing	High

1.4.2 Troubleshooting Procedure — Pressure Fluctuation

1. Check upstream suction pressure stability — multistage pumps are highly sensitive to suction conditions.
2. Inspect minimum-flow recirculation valve for correct operation; a stuck-open recirculation valve causes internal recirculation and pressure instability.
3. Review trend data for the past 24 hours; gradual fluctuation growth suggests internal wear, sudden onset suggests a control system or valve issue.
4. Any visible hydrocarbon seepage requires immediate isolation per facility hydrocarbon release procedure — this is a Critical safety event regardless of pressure readings.

1.4.3 Required Parts

- Stage wear ring kit (Part No. WR-318-5ST)
- Flange gasket set, API 6A spec (Part No. GSK-318-API6A)

1.4.4 Escalation Rule

Any mention of **crude oil leak**, **hydrocarbon smell**, or **seepage** on a P-318 ticket is an automatic Critical classification — this equipment handles high-pressure hydrocarbons and even minor seepage is treated as a process safety event under facility HSE policy.

SECTION 2: RECIPROCATING & SCREW COMPRESSORS

2.1 Equipment Overview — Compressor Class

Compressors increase gas pressure for transport, processing, or injection. Reciprocating compressors use pistons and are common for high-pressure, lower-volume duties; screw compressors use interlocking rotors and suit continuous medium-pressure service. Valve failure is the single most common reciprocating compressor failure mode industry-wide, and liquid carryover into the cylinder is the most dangerous.

2.2 Equipment Record: Compressor C-11

Equipment Type: Reciprocating Compressor, 2-Stage
Service: Gas Lift Injection
Rated Capacity: 12 MMSCFD
Discharge Pressure: 2,400 psig
Driver: Gas Engine, 1,800 HP

2.2.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Compressor stopped, production halted	Trip event (multiple possible causes)	Critical — production impact
Knocking sound from cylinder	Valve failure or liquid carryover	Critical — risk of catastrophic damage
Discharge temperature trending upward over hours	Valve leakage (internal recirculation)	Medium-High
Unstable suction pressure	Upstream scrubber level issue	High — check for liquid carryover risk
Excessive vibration at skid base	Foundation bolt loosening or piping resonance	Medium

2.2.2 Troubleshooting Procedure — Compressor Trip / Production Halt

1. Check the trip log on the local control panel — identify the FIRST trip cause logged (subsequent alarms are often secondary effects, not root cause).
2. Common trip causes for this unit, in order of frequency: high discharge temperature, low lube oil pressure, high vibration, suction scrubber high level.
3. If trip cause is **high scrubber level**, do NOT attempt restart until scrubber is confirmed drained — liquid carryover into a reciprocating compressor cylinder can cause immediate, severe mechanical damage and is a safety hazard due to potential cylinder over-pressurization.
4. If trip cause is **low lube oil pressure**, inspect oil pump and filter before any restart attempt.
5. Production-halting trips on gas lift compressors are escalated to the Production Supervisor in addition to standard maintenance ticket routing — gas lift loss can affect well productivity within hours.

2.2.3 Required Parts (Valve Replacement)

- Suction/discharge valve set, Stage 1 (Part No. VLV-C11-S1)
- Suction/discharge valve set, Stage 2 (Part No. VLV-C11-S2)
- Cylinder gasket kit (Part No. GSK-C11-CYL)
- Estimated downtime: 8-12 hours for full valve change-out

2.2.4 Escalation Rule

Any C-11 ticket mentioning **stopped**, **halted**, or **production impact** is classified Critical — gas lift compressor downtime directly affects well production and triggers automatic escalation per facility production-continuity policy, independent of the underlying mechanical cause.

2.3 Equipment Record: Compressor C-22

Equipment Type: Oil-Flooded Rotary Screw Compressor
Service: Plant/Instrument Air Supply
Rated Capacity: 850 SCFM
Discharge Pressure: 125 psig

2.3.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Air pressure dropping plant-wide	Compressor underperforming or tripped	High — affects multiple instrument air users
High oil carryover in discharge air	Oil separator element saturated	Medium — schedule element replacement
Compressor cycling on/off rapidly (short-cycling)	Pressure switch fault or air demand mismatch	Medium
Unusual screech/whine on startup	Belt slippage or bearing wear	Medium-High

2.3.2 Troubleshooting Procedure — Plant Air Pressure Loss

1. Check whether C-22 is running and at what load percentage on the local HMI.
2. Instrument air pressure loss is treated with elevated urgency because pneumatically-actuated safety valves across the plant rely on instrument air to fail safely — sustained low air pressure can compromise multiple safety systems simultaneously.
3. If C-22 has tripped, check trip log; common causes are high discharge temperature and oil separator differential pressure.
4. If backup compressor C-22B is available, confirm auto-changeover has engaged; if not, manually start backup unit before further diagnosis on C-22.

2.3.3 Required Parts

- Oil separator element (Part No. SEP-C22-EL4)
- Drive belt set (Part No. BLT-C22-V5)
- Air filter element (Part No. FLT-C22-AF2)

2.3.4 Escalation Rule

Plant-wide instrument air pressure loss is escalated to High urgency at minimum, even without an explicit safety keyword, because of its facility-wide impact on pneumatic safety systems — this is a documented exception to standard single-equipment severity scoring.

2.4 Equipment Record: Compressor C-305

Equipment Type: Centrifugal Compressor, Single-Stage
Service: Vapor Recovery
Rated Capacity: 6.5 MMSCFD
Driver: Electric Motor, 950 HP

2.4.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Surge alarm active	Operating point near surge line	Critical — risk of mechanical damage from surge cycling
Gas detector alarm near compressor skid	Possible seal gas leak	Critical — toxic/flammable vapor exposure risk
Bearing temperature trending up slowly	Lubrication degradation	Medium
Anti-surge valve not responding	Control valve actuator fault	High — surge protection compromised

2.4.2 Troubleshooting Procedure — Surge Alarm

1. Surge events on centrifugal compressors can cause severe thrust bearing damage within seconds of repeated cycling — this is one of the few failure modes where seconds matter for equipment protection, not just hours.
2. Verify anti-surge valve is responding to the control signal; if not, this is the most likely root cause and requires immediate maintenance attention.
3. Do not attempt to manually override anti-surge protection logic under any circumstances without Operations Engineering sign-off.
4. If gas detection has also alarmed near this skid (vapor recovery handles flammable hydrocarbon vapors), treat as a combined mechanical/safety event — evacuate non-essential personnel from the immediate area per facility gas alarm response procedure.

2.4.3 Required Parts

- Anti-surge valve actuator rebuild kit (Part No. ASV-C305-RBK)
- Dry gas seal cartridge (Part No. DGS-C305-CART)

2.4.4 Escalation Rule

Any C-305 ticket combining **surge** with **gas detector** or **gas alarm** language is automatically Critical — this combination indicates both a mechanical protection failure and a potential flammable vapor release, the two highest-priority risk categories for this equipment class.

SECTION 3: ELECTRIC MOTORS & DRIVES

3.1 Equipment Overview — Electric Motor Class

Electric motors drive the majority of rotating equipment in the facility (pumps, compressors, fans). Failure modes split into electrical (winding insulation breakdown, bearing currents) and mechanical (bearing wear, misalignment with driven equipment, cooling fan failure). Thermal stress is the leading cause of premature motor failure across the industry.

3.2 Equipment Record: Motor M-18

Equipment Type: Induction Motor, TEFC
Service: Driver for Pump P-204 (see Section 1.2)
Rated Power: 250 kW
Rated Speed: 2,980 RPM
Voltage: 6.6 kV

3.2.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Temperature slightly above normal, no other symptoms	Cooling fan partial blockage or ambient temperature rise	Low — monitor trend
Temperature significantly above normal, sustained	Winding insulation stress or bearing friction	Medium-High
Temperature alarm with burning smell	Insulation failure in progress	Critical — risk of winding fire
Motor trips on overload repeatedly	Mechanical binding in driven pump, or motor winding fault	High — investigate both motor and driven equipment
Unusual humming/buzzing at startup	Voltage imbalance across phases	Medium

3.2.2 Troubleshooting Procedure — Elevated Temperature

1. Compare current winding temperature reading (via embedded RTD) against the equipment's normal operating baseline, not against a generic threshold — baselines vary by ambient conditions and loading.
2. Check cooling fan operation and air intake/exhaust louvers for blockage (dust, debris, nesting insects are common culprits in this climate).
3. Verify driven pump (P-204) is not experiencing a mechanical issue that increases motor load — elevated motor temperature is sometimes a symptom of a pump-side problem, not a motor fault.
4. A "slightly higher than normal" temperature reading with no other symptoms typically warrants Low to Medium urgency and scheduled monitoring rather than immediate intervention — avoid over-escalating minor thermal trend deviations that lack corroborating symptoms.
5. Any burning smell, smoke, or visible discoloration of the motor casing escalates this immediately to Critical regardless of the numeric temperature reading.

3.2.3 Required Parts

- Cooling fan assembly (Part No. FAN-M18-TEFC)
- Bearing set (Part No. BRG-M18-6.6KV)
- RTD sensor replacement (Part No. RTD-M18-PT100)

3.2.4 Escalation Rule

A temperature-related M-18 ticket is escalated to Critical ONLY if it includes burning smell, smoke, or visible damage. A standalone "slightly higher than normal" temperature reading, with no other symptoms, should generally be classified Low or Medium — this is a deliberate calibration point in the knowledge base to test whether the classification agent over-escalates mild, non-corroborated symptoms.

3.3 Equipment Record: Motor M-44

Equipment Type: Induction Motor, TEFC
Service: Driver for Pump P-207 (see Section 1.3)
Rated Power: 400 kW
Rated Speed: 1,490 RPM
Voltage: 6.6 kV

3.3.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Vibration at motor bearing housing	Coupling misalignment with P-207	Medium
Motor fails to start, breaker trips immediately	Short circuit or severe winding fault	Critical — do not attempt repeated restarts
Intermittent tripping under normal load	Loose electrical connection at terminal box	High — fire risk from arcing

3.3.2 Troubleshooting Procedure — Failure to Start

1. Do NOT attempt more than one restart after an immediate breaker trip — repeated restart attempts on a genuine short-circuit fault risk equipment damage and arc-flash hazard to personnel.
2. Check terminal box for loose connections, discoloration, or burn marks before any restart attempt.
3. If breaker trips instantly on closing (before motor draws significant current), this indicates a likely short circuit — isolate and lock out for electrical inspection.

3.3.3 Required Parts

- Terminal box connection kit (Part No. TB-M44-CONN)
- Motor winding insulation test and repair (specialist contractor required for winding rewind)

3.4 Equipment Record: Motor M-091

Equipment Type: Induction Motor, Explosion-Proof
Service: Driver for Pump P-318 (see Section 1.4)

Rated Power: 1,100 kW

Rated Speed: 2,980 RPM

Voltage: 11 kV

3.4.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Gland/seal area showing condensation	Normal in humid conditions, monitor only	Low
Explosion-proof enclosure showing any crack or gap	Loss of explosion-proof rating	Critical — immediate isolation required
High-pitched electrical whine	Bearing currents from VFD harmonics	Medium

3.4.2 Troubleshooting Procedure — Enclosure Integrity

1. This motor is rated explosion-proof (Ex d) due to its location in a hydrocarbon-classified hazardous area. ANY visible crack, gap, or damage to the enclosure flame path is treated as Critical, regardless of whether the motor is currently operating normally electrically.
2. Do not operate this motor with a compromised enclosure under any circumstances — this is a zone classification violation, not a performance issue.
3. Standard thermal/vibration troubleshooting (as per M-18, Section 3.2.2) applies for non-enclosure-related symptoms.

3.4.3 Escalation Rule

Any M-091 ticket mentioning **crack**, **damaged enclosure**, **gap**, or **explosion-proof rating** is automatically Critical — this is a hazardous-area equipment integrity issue, treated with the same priority as a direct gas leak.

SECTION 4: STEAM & GAS TURBINES

4.1 Equipment Overview — Turbine Class

Turbines convert thermal/kinetic energy in steam or combustion gas into rotational mechanical energy. They are typically the highest-value rotating equipment on site and failures carry significant production and safety consequences. Turbine incidents are escalated with a lower symptom threshold than pumps or motors due to the energy involved in a turbine failure event.

4.2 Equipment Record: Turbine T-501

Equipment Type: Single-Stage Steam Turbine
Service: Mechanical Drive for Compressor C-305 (see Section 2.4)
Rated Power: 950 kW
Rated Speed: 8,200 RPM

4.2.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Overspeed trip activated	Governor control fault or load loss	Critical — verify trip system functioned correctly before any restart
Steam leak at gland seal	Seal wear	High — steam at this pressure causes severe burns
Axial displacement alarm	Thrust bearing wear	Critical — risk of rotor-to-stator contact
Casing vibration trending upward	Blade fouling, erosion, or bearing wear	Medium-High depending on rate of change

4.2.2 Troubleshooting Procedure — Overspeed Trip

1. An overspeed trip is a protective system functioning as designed — the priority is confirming WHY overspeed occurred, not simply resetting and restarting.
2. Check for sudden loss of load on driven compressor C-305 (e.g., a trip on C-305 itself can cause T-501 to overspeed as load is suddenly removed).
3. Verify the mechanical overspeed trip mechanism reset correctly and test electronic overspeed protection before returning to service.
4. Turbine overspeed events are NEVER closed out as routine — facility policy requires a documented root-cause review before restart authorization, given the catastrophic failure potential of rotor overspeed.

4.2.3 Required Parts

- Gland seal packing set (Part No. GLD-T501-PK3)
- Governor valve actuator (Part No. GOV-T501-ACT, long lead time — 3-4 weeks)

4.2.4 Escalation Rule

ALL T-501 tickets are classified at minimum High urgency by default, even for seemingly minor symptoms — this turbine drives a critical vapor recovery compressor and operates at high speed/high energy, so the facility's risk tolerance for "wait and see" is lower than for slower rotating equipment.

4.3 Equipment Record: Turbine T-512

Equipment Type: Gas Turbine, Industrial Twin-Shaft

Service: Power Generation

Rated Power: 25 MW

4.3.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Exhaust gas temperature spread alarm	Combustor or fuel nozzle imbalance	High — risk of hot section damage
Compressor surge/stall on startup	Inlet guide vane scheduling fault	High
Vibration alarm on accessory gearbox	Gearbox bearing wear	Medium-High
Flame detector fault during startup sequence	Sensor fault OR genuine flame-out condition	Critical — must be verified, never bypassed

4.3.2 Troubleshooting Procedure — Flame Detector Fault

1. Never bypass a flame detector fault to force a startup sequence to continue — this safety interlock exists specifically to prevent unburned fuel accumulation in the combustor, which can cause a violent ignition event ("hot startup" / explosion) if bypassed.
2. Confirm whether the fault is a genuine flame-out (check for actual ignition confirmation from a second, independent flame detector if fitted) or a sensor-level fault.
3. This is one of the few interlocks on site where the documented procedure is: troubleshoot in place, do NOT attempt a workaround restart, even under production pressure.

4.3.3 Escalation Rule

Any T-512 ticket mentioning **flame detector**, **flame-out**, or **failed ignition** is Critical, full stop, regardless of any other context in the ticket — this matches the facility's hard safety-interlock policy for gas turbine startup systems.

SECTION 5: VALVES & PRESSURE RELIEF SYSTEMS

5.1 Equipment Overview — Valve Class

Valves control flow, isolate equipment, and protect against overpressure. Pressure relief valves (PRVs) are the last line of defense against vessel/pipe overpressure and receive the strictest handling procedures of any equipment class on site — incorrect handling of a PRV ticket can directly defeat a critical safety barrier.

5.2 Equipment Record: Valve V-118

Equipment Type: Pressure Relief Valve
Service: Overpressure protection for Separator Vessel SEP-09
Set Pressure: 18 barg

5.2.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Valve found "simmering" (slight leakage below set pressure)	Seat wear or fouling	High — relief capacity may be compromised
Valve lifted and reseated normally during an overpressure event	Valve functioned as designed	Medium — inspect and re-certify before next service interval if due
Valve fails to reseat after lifting	Seat damage or debris	Critical — vessel left without overpressure protection AND continuous venting
Tamper-evident seal broken without authorization	Possible unauthorized adjustment	Critical — safety integrity cannot be confirmed until inspected

5.2.2 Troubleshooting Procedure — Valve Fails to Reseat

1. A relief valve that fails to reseat leaves the protected vessel (SEP-09) without overpressure protection — this is treated as equivalent in severity to having no relief valve installed at all, regardless of current vessel pressure.
2. Do not attempt field adjustment of a pressure relief valve under any circumstances — these are precision safety devices and unauthorized adjustment voids certification and may violate regulatory requirements.
3. If continuous venting is occurring, assess the vented fluid for hazard classification (flammable, toxic) and follow facility gas/vapor release procedure in parallel with the mechanical response.
4. Replacement or bench-testing must be performed by a certified relief valve technician — this is not a standard maintenance task.

5.2.3 Escalation Rule

ANY ticket regarding V-118 (or any pressure relief valve) failing to reseat, leaking continuously, or showing signs of tampering is automatically Critical — relief valves are the last layer of overpressure protection and any compromise to their function is treated as a process safety event, not a routine mechanical fault.

5.3 Equipment Record: Valve V-203

Equipment Type: Control Valve, Globe Pattern
Service: Flow control on crude feed to distillation unit
Actuator Type: Pneumatic, fail-closed

5.3.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Valve not responding to control signal	Actuator diaphragm failure or air supply loss	High — loss of flow control
Valve hunting (oscillating position)	Tuning issue or positioner fault	Medium
Visible leakage at valve stem packing	Packing wear	Medium-High depending on process fluid

5.3.2 Troubleshooting Procedure — Loss of Response

1. Confirm instrument air supply pressure to the actuator first — many "valve faults" are actually upstream air supply issues (see Section 2.3 on plant air compressor C-22).
2. This valve is fail-closed by design — if it has failed and closed, crude feed flow has stopped, which should be cross-checked against process data before assuming a valve-only issue.
3. Check positioner feedback signal against actual valve position using a handheld calibrator.

5.3.3 Required Parts

- Actuator diaphragm kit (Part No. ACT-V203-DIA)
- Stem packing set (Part No. PKG-V203-ST2)

SECTION 6: HEAT EXCHANGERS

6.1 Equipment Overview — Heat Exchanger Class

Heat exchangers transfer thermal energy between process streams without direct mixing. As static (non-rotating) equipment, failures are typically gradual (fouling, fin damage) rather than sudden — with the notable exception of tube rupture, which is a high-severity event regardless of how gradually it develops.

6.2 Equipment Record: Heat Exchanger HX-77

Equipment Type: Shell-and-Tube Heat Exchanger
Service: Crude Oil Preheat
Design Pressure (Shell/Tube): 25 barg / 40 barg

6.2.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Outlet temperature gradually declining over days/weeks	Fouling buildup on tube side	Low-Medium — schedule cleaning
Sudden pressure equalization between shell and tube side	Possible tube rupture	Critical — cross-contamination risk
External shell leak	Gasket or shell corrosion	High — hydrocarbon release risk if in crude service
Increased pressure drop across tube bundle	Partial blockage or fouling	Medium

6.2.2 Troubleshooting Procedure — Suspected Tube Rupture

1. A sudden, unexplained pressure equalization between the shell side and tube side is the primary indicator of tube rupture — this allows uncontrolled cross-contamination between the two process streams and is treated as Critical even before visual confirmation.
2. Isolate the exchanger per facility lockout procedure immediately upon suspected tube rupture — do not wait for confirmation before initiating isolation, given the contamination and potential overpressure risk to the lower-design-pressure side.
3. Gradual fouling-related performance decline (the much more common scenario) does NOT require emergency isolation and should be scheduled as routine cleaning during next planned maintenance window.

6.2.3 Escalation Rule

The distinction between "gradual performance decline" (Low-Medium, routine) and "sudden pressure/temperature anomaly" (Critical, suspected rupture) is the single most important classification judgment for this equipment — tickets must be read carefully for words like "sudden," "rupture," or unexplained cross-stream pressure changes versus gradual trend language like "slowly," "over the past week," or "gradually."

6.3 Equipment Record: Heat Exchanger HX-88

Equipment Type: Plate Heat Exchanger
Service: Cooling Water / Lube Oil Cooling
Design Pressure: 16 barg

6.3.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Lube oil temperature trending up	Fouling on cooling water side, or reduced cooling water flow	Medium-High — affects bearing lubrication elsewhere
Visible external leak at plate pack gasket	Gasket degradation	Medium — typically low-pressure, contained leak
Cooling water discoloration at outlet	Possible internal cross-contamination	High — investigate for internal leak between circuits

6.3.2 Troubleshooting Procedure — Rising Lube Oil Temperature

1. Check cooling water supply flow and temperature first — this exchanger's performance is entirely dependent on adequate cooling water flow.
2. Note that this exchanger often serves a downstream rotating equipment item's lubrication system — rising oil temperature here can become a secondary cause of bearing issues elsewhere (cross-reference with the specific pump/compressor/motor this unit cools, per the cooling water distribution diagram).

SECTION 7: ELECTRICAL SWITCHGEAR & MOTOR CONTROL CENTERS

7.1 Equipment Overview — Electrical Distribution Class

Switchgear and Motor Control Centers (MCCs) distribute and protect electrical power to rotating equipment and facility systems. Faults here carry elevated risk due to arc-flash hazard potential and the cascading impact of a single failure on multiple downstream equipment items.

7.2 Equipment Record: Motor Control Center MCC-14

Equipment Type: Low Voltage MCC (415V)
Service: Feeds multiple utility pumps and instrument air compressors
Number of Feeder Sections: 12

7.2.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
Single feeder breaker tripped, others normal	Overload or fault on that specific downstream load	Medium-High depending on what it feeds
Burning smell from MCC room	Insulation or connection fault in progress	Critical — potential fire, evacuate and de-energize
Visible arcing or flash at any breaker	Active electrical fault	Critical — extreme danger, do not approach, evacuate immediately
Multiple feeders tripped simultaneously	Upstream supply fault or bus fault	Critical — facility-wide impact likely

7.2.2 Troubleshooting Procedure — Burning Smell / Suspected Fire

1. A burning smell or any visible arcing in an MCC room is one of the highest-priority safety events covered by this manual — electrical fires in MCC rooms can spread rapidly and pose severe arc-flash and toxic-smoke risk to personnel.
2. Evacuate the immediate area and notify the fire/emergency response team per facility procedure BEFORE attempting any investigation — do not enter an MCC room with a burning smell to "take a look" without following lockout/evacuation procedure first.
3. Only qualified, PPE-equipped electrical personnel may approach an energized MCC suspected of an internal fault — this is explicitly not a general maintenance technician task.

7.2.3 Escalation Rule

Any MCC-14 ticket mentioning **burning smell**, **smoke**, **arcing**, or **flash** is automatically Critical and is routed to both the maintenance system AND the facility emergency response notification list simultaneously — this is one of the only equipment types in this manual with dual-channel escalation.

7.3 Equipment Record: Switchgear SW-29

Equipment Type: Medium Voltage Switchgear (11kV)

Service: Primary distribution to Motor M-091 (see Section 3.4) and adjacent feeders

7.3.1 Common Fault Signatures

Symptom	Likely Cause	Severity Indicator
SF6 gas pressure alarm (if gas-insulated)	Gas leak from compartment seal	High — insulation integrity at risk
Partial discharge monitoring alarm	Insulation degradation, early stage	Medium — schedule inspection
Breaker fails to close on command	Mechanism fault or control circuit issue	High — affects ability to restore power after any trip
Unexplained nuisance tripping	Protection relay setting issue or genuine intermittent fault	Medium-High — requires careful investigation, don't dismiss as "nuisance" without data review

7.3.2 Troubleshooting Procedure — Breaker Fails to Close

1. Medium voltage switchgear faults require certified high-voltage electrical personnel — do not attempt manual intervention beyond visual inspection from a safe distance.
2. Check control circuit voltage and breaker spring-charge status (many MV breakers require a charged spring mechanism to close, separate from the protection/trip circuit).
3. A breaker that fails to close after a protective trip should NOT be force-closed or bypassed — if it tripped for a valid protection reason, forcing closure can result in repeat fault energization with full short-circuit current.

7.3.3 Escalation Rule

SW-29 tickets default to at least High urgency given this switchgear's role feeding the explosion-proof motor M-091 in a hazardous area — any extended loss of this feeder has downstream safety implications beyond the switchgear itself.

SECTION 8: GENERAL SAFETY OVERRIDE REFERENCE

8.1 Purpose of This Section

This section consolidates the facility-wide, equipment-independent safety override triggers referenced throughout Sections 1-7. These are the same categories of keywords that `backend/services/safety_rules.py` checks for programmatically — this section exists in the knowledge base so that retrieval and drafting agents can cite a consistent, authoritative source when explaining WHY a safety override applied, rather than the override appearing to come from nowhere.

8.2 Universal Critical-Override Triggers

The following conditions force Critical classification regardless of equipment type, and regardless of what the classification model itself concludes from context:

Trigger Category	Example Phrases	Rationale
Uncontrolled hydrocarbon release	"leak," "seepage," "spill" (in hydrocarbon service)	Fire/explosion and environmental risk
Toxic/flammable gas detection	"gas leak," "H2S," "gas detector alarm"	Acute health and explosion risk
Fire indicators	"fire," "smoke," "burning smell"	Immediate life-safety risk
Explosion-proof integrity loss	"cracked enclosure," "damaged Ex rating"	Hazardous area ignition risk
Safety system failure	"relief valve failed to reseat," "flame detector bypassed"	Loss of last-line-of-defense protection
Electrical arc/fire	"arcing," "flash," "electrical fire"	Arc-flash and fire risk to personnel
Structural integrity loss	"structural damage," "crack in pressure boundary"	Catastrophic failure risk

8.3 Deliberately NON-Critical Reference Cases

To properly calibrate classification (and avoid a system that over-escalates everything to Critical, which would be just as useless as under-escalating), this manual includes several scenarios that should NOT trigger Critical classification despite sounding concerning at first read:

- **Motor M-18 "temperature slightly higher than normal" with no other symptoms** (Section 3.2.4) — Low/Medium, not Critical.
- **Pump P-207 strainer blockage causing reduced flow** (Section 1.3.1) — Low/Medium, routine.
- **Heat exchanger HX-77 gradual fouling-related performance decline** (Section 6.2.3) — Low/Medium, routine, NOT the same as sudden tube rupture.
- **Steam turbine T-501 minor gland seal leak, no other symptoms** — High but not necessarily Critical (distinct from an axial displacement or overspeed event).

8.4 Notes for Drafting Agent Citation

When generating a recommendation, the drafting agent should explicitly state which section and rule justified its classification — e.g., "Per Section 1.2.4, this ticket's mention of leakage on Pump P-204 escalates urgency to Critical." This traceability is the entire point of grounding generation in retrieved manual content rather than allowing the model to assert urgency without citing a source.

End of TF-MAN-001, Revision 2.1. This document is synthetic training data for the TriageFlow RAG knowledge base and does not represent any real facility, equipment, or incident.